

SU University – React Practical Examination

Time: 90 Minutes

Task 01: React Student Portal Application

Instructions:

- Use functional components only.
 - You may use the internet and AI tools.
 - Use React Router.
 - Use Hooks properly.
 - Write clean and structured code.
 - Basic styling is required.
 - Do not refresh the page for navigation.
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Objective

Build a React application with routing, API integration, form validation, localStorage usage, and state management.

Application Requirements

You must create 4 routes:

1. / → Home
2. /students → Students (API Page)
3. /add → Add Student
4. /counter → Counter

A Navbar must be visible on all pages.

1. Navbar (Mandatory)
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Use:

- BrowserRouter
- Routes
- Route
- NavLink (preferred)

Navbar must contain links to:

- Home
- Students
- Add Student
- Counter

Active link should be visually different.

2. Home Page (/)

Display:

- Title: "Student Portal"
- Total number of students stored in localStorage

Behavior:

- On page load, read students from localStorage.
- If no students exist, display:
"No students added yet."

Hooks Required:

- useState
- useEffect

3. Students Page (/students)

API URL:

<https://jsonplaceholder.typicode.com/users>

Requirements:

- Fetch data using fetch()
- Use useEffect
- Show "Loading..." while fetching
- Show error message if API fails
- Display only the first 6 users

Each user card must show:

- Name
- Email
- Phone

Display users in card format using proper CSS styling.

Hooks Required:

- useState
- useEffect

4. Add Student Page (/add)

Create a form with the following fields:

- Name (required)
- Email (required)
- Phone (required)
- Gender (radio button: Male/Female)
- Submit Button

Validation Rules:

- Name cannot be empty
- Email must contain "@"
- Phone must be exactly 10 digits
- Show error messages below respective fields

On Submit:

If validation fails:

- Show appropriate error messages

If validation succeeds:

- Store the student object inside localStorage (store as an array of students)
- Show a success message
- Clear the form fields

Bonus (Optional but Recommended):

- Display the newly added student below the form immediately.

Hooks Required:

- useState

5. Counter Page (/counter)

Create a functional component named Counter.

Initial State:

value = 0

Functions:

- Increment → Increase by 1
- Decrement → Decrease by 1 (value must not go below 0)
- Reset → Set value back to 0

UI Requirements:

- Display the current counter value
- Add three buttons: Increment, Decrement, Reset

Logic Rule:

- Prevent counter from going below 0

Color Rule:

- Show value in red when value = 0
- Show value in green when value > 0

Hook Required:

Here is the extracted text from the image:

useState
Technical Constraints
Functional components only
Must use:
useState
useEffect
Must use React Router
Must use localStorage
Basic CSS styling required
Application must not crash